

Birla Institute of Technology & Science, Pilani, Rajasthan
First Semester 2025-2026
Tutorial-8

Course: ECE/EEE F311 Communication Systems
Instructor-in-Charge: Prof. Syed Mohammad Zafaruddin

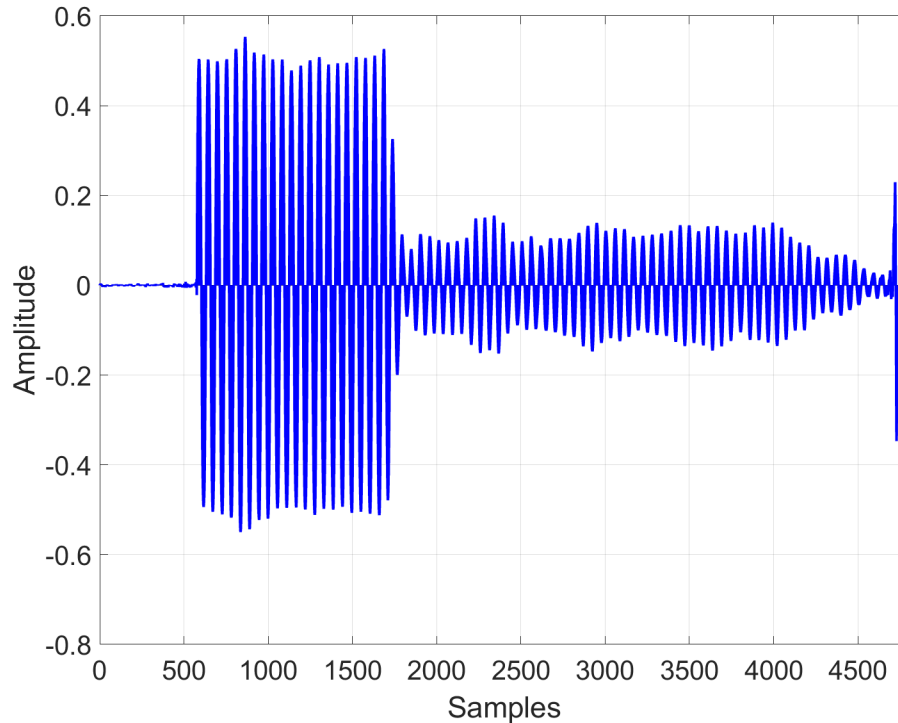
Date: 01-10-2025

Objectives

- PCM, DM, Nyquist Criteria

Tutorial Problems

1. An audio clip of duration 1 second is shown in figure. Assume average power of signal as 2 Watt and that the amplitude has signal dynamics of ± 0.6 Volt.



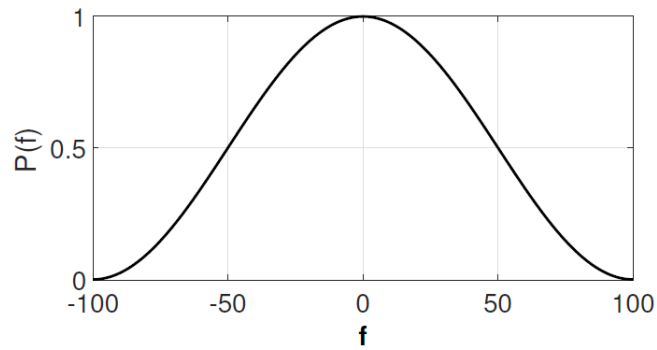
- (a) Determine the specification (i.e., number of bits/sample) of analog-to-digital converter (ADC) using PCM to achieve an SQNR of 60 dB.
- (b) Determine the minimum channel bandwidth required to transmit the audio clip with 8-ary transmission.

2. Problem from Text Book

6.7-1 In a single-integration DM system, the voice signal is sampled at a rate of 64 kHz, similar to PCM. The maximum signal amplitude is normalized as $A_{\max} = 1$.

- (a) Determine the minimum value of the step size σ to avoid slope overload.
- (b) Determine the granular noise power N_o if the voice signal bandwidth is 3.4 kHz.
- (c) Assuming that the voice signal is sinusoidal, determine S_o and the SNR.
- (d) Assuming that the voice signal amplitude is uniformly distributed in the range $(-1, 1)$, determine S_o and the SNR.
- (e) Determine the minimum transmission bandwidth.

3. Use the Nyquist frequency domain $\sum_{k=-\infty}^{\infty} P(f + kR_s) = \text{constant}$ to show that the frequency spectrum $P(f)$ of a raised cosine pulse (as shown in the figure) satisfies the Nyquist criterion for zero ISI. Find $p(t)$.



Solution

1. (a)

$$\text{SQNR}_{\text{linear}} = \frac{3L^2 P_m}{m_p^2}$$

Solving $L = 244.95$; $L = 2^N$ gives $N = 7.94$, rounded $N = 8$.

(b) From figure, assume sampling rate 4600 samples (You can take close to 4800 too as per graph) per second. $R_b = 8 \times 4600$ bps

$$R_s = \frac{8 \times 4600}{3} = 12266 \text{ symbols/sec}$$

$$B = R_s/2 = 6.133 \text{ KHz}$$

2. **Solution 4:** BP Lathi 5.7-1: Assume sinusoidal signal approximation for voice signal:

(a) $A = \frac{\Delta f_s}{2\pi f_m}$; Take $f_m = 800$ Hz. This gives $\Delta = 0.0785$

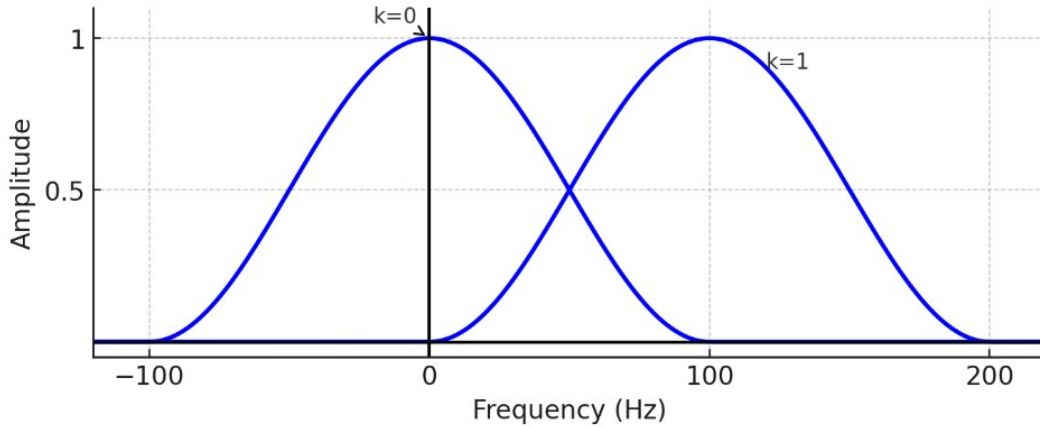
(b) $N_0 = \frac{\Delta^2 B}{3f_s} = 1.12 \times 10^{-4}$

(c) $S_0 = \frac{A^2}{2} = 0.5$; $\text{SNR} = \frac{0.5}{1.12 \times 10^{-4}} = 4.46 \times 10^3$

(d) For uniform distribution: $S_0 = \frac{m_p^2}{3}$; $\text{SNR} = \frac{0.33}{1.12 \times 10^{-4}} = 2.94 \times 10^3$

(e) BW = half of data rate = 32 KHz.

3. From the figure at $k = 0$ and $k = 1$;



$$\sum_{k=-\infty}^{\infty} P(f + kR_s) = \text{constant} \Rightarrow R_s = 100,$$

$$\text{BW} = 100 \text{ Hz},$$

$$\text{BW} = \frac{(\alpha + 1)R_s}{2} \Rightarrow \alpha = 1 \quad (\text{full raised-cosine}),$$

$$p(t) = R_s \frac{\cos(\pi R_s t)}{1 - 4R_s^2 t^2} \text{sinc}(R_s t),$$

$$p(t)|_{R_s=100} = 100 \frac{\cos(100\pi t)}{1 - 40000 t^2} \text{sinc}(100t),$$